

① Calculate the energy, momentum and de-Broglie's wavelength for an electron trapped in a one-dimensional quantum well of size $10\text{\AA}$ in its ground state

$$E_n = \frac{n^2 h^2}{8mL^2}$$

$$n=1, E_1 = \frac{h^2}{8mL^2} = \frac{6.63 \times 6.63 \times 10^{-68}}{8 \times 9.1 \times 10^{-31} \times 10^{-18}}$$

$$E_1 = 0.6038 \times 10^{-19} \text{ J} = 0.377 \text{ eV}$$

$$p = \sqrt{2mE} = \frac{(\sqrt{2m}) n h}{(2)(\sqrt{2})(m)(L)}$$

$$p = \frac{h}{2L} = \frac{6.63 \times 10^{-34}}{2 \times 10^{-9}} = 3.315 \times 10^{-25} \text{ kg m/sec}$$

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{3.315 \times 10^{-25}} = 2 \times 10^{-9} \text{ m}$$

② Calculate the difference in energies of first two allowed/permitted states for an electron confined to a strip of 1 nm .

$$L = 10^{-9} \text{ m}$$

$$E_1 = \frac{h^2}{8mL^2}, \quad E_2 = \frac{4h^2}{8mL^2}$$

$$E_2 - E_1 = \frac{3h^2}{8mL^2} = \frac{3 \times 6.63 \times 6.63 \times 10^{-68}}{8 \times 9.1 \times 10^{-31} \times 10^{-18}}$$

$$= 1.81 \times 10^{-19} \text{ J}$$

$$= 1.13 \text{ eV}$$

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- ③ Consider a marble of mass 10 gm confined to box of size 1 cm. Assuming the concept of particle in box, determine its energy for first two permitted states and its difference. Comment on the answer.

$$L = 10^{-2} \text{ m}, \quad m = 10 \times 10^{-3} \text{ kg}$$

$$E_n = \frac{n^2 h^2}{8mL^2}$$

$$E_1 = \frac{h^2}{8mL^2} = \frac{43.9569 \times 10^{-68}}{8 \times 10^{-2} \times 10^{-4}}$$

$$E_1 = 5.49 \times 10^{-62} \text{ J}$$

$$E_2 = 21.97 \times 10^{-62} \text{ J}$$

$$E_2 - E_1 = 16.48 \times 10^{-62} \text{ J}$$

difference is very small and hence can be considered as continuous. There is no energy quantization.

- ④ An electron is trapped in one dim pot. well of length 0.1 nm. Calculate the energy required to excite from ground state to third excited state

$$n=1, \quad n=4$$

$$E_1 = 6.04 \times 10^{-18} \text{ J}$$

$$E_4 - E_1 = 90.6 \times 10^{-18} \text{ J}$$

$$E_4 = 96.64 \times 10^{-18}$$

$$= 56.625 \times 10^{-18} \times 10^{+19}$$

$$= \underline{\underline{566.25 \text{ eV}}}$$

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